Signal-Conditioned Memory Compression for Multi-Turn Conversations

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Abstract

Compressing long multi-turn conversations requires deciding not only how to compress context, but which signal should rank memory candidates under a fixed budget. We study a sparse keep/compress/drop (KCD) codec and compare geometric, semantic, and hybrid scoring signals across conversational memory regimes. On a stress-test benchmark designed to expose support-turn failures, a geometry-scored codec substantially outperforms a semantic signal swap inside the same codec. On MSC (1,000 conversations), we show that geometry-based compression methods substantially outperform LongLLMLingua on geometric fidelity (ΔL2 $\approx$ $+130$) while LongLLMLingua modestly improves answer NLL—a geometry-behavior divergence confirming that behavioral metrics alone are insufficient to evaluate memory compression. Among KCD signal variants on MSC (1,000 conversations, Qwen2.5-1.5B), query-conditioned geometry achieves the best combination of geometric fidelity and behavioral quality simultaneously: at budget 0.50 it reduces logit L2 by 109.4 units versus semantic KCD ($p < 0.001$, conversation-level) while also achieving best answer NLL across all budgets. On LongMemEval-S (100 full-length conversations), all KCD variants significantly improve behavioral quality over uniform compression. A Llama-3.2-3B cross-model check shows that the geometry signal advantage is model-family-dependent: semantic compression dominates geometry compression on behavior across all budgets for Llama-3B ($p \leq 0.01$, conversation-level). The paper's central claim is that geometry-behavior divergence is real and consequential, and that query-conditioned geometry is the signal that best resolves it on Qwen2.5 models.

1 Introduction

Long conversations create a memory-allocation problem before they create a generation problem. Once the full dialogue no longer fits inside a practical inference budget, a system must decide which parts of the conversation to retain verbatim, which parts to compress into sparse surrogates, and which parts to evict. This is not only a capacity problem: even when the full context fits, language models fail to uniformly exploit information distributed across long histories—performance degrades substantially for content appearing in the middle of the context window (Liu et al., 2024).

Existing approaches to managing this include retrieval augmentation (Lewis et al., 2020), hierarchical memory management (Packer et al., 2024), prompt compression (Jiang et al., 2024), and token-level cache heuristics (Zhang et al., 2023; Li et al., 2024). All are useful, but none directly address a central question for *in-context* conversational memory compression: *what signal should determine memory importance under budget?* We focus on inference-time compression that operates within the context window without external indexes or model modifications.

This paper studies that question through a sparse keep/compress/drop (KCD) codec for conversational memory. The codec is deliberately simple. Conversation segments can be kept in full, compressed into anchor-plus-support summaries, or evicted. The variable of interest is the *scoring signal* that drives those actions. We compare three regimes: geometry-based scoring from the conversation's hidden-state trajectory, semantic scoring from query similarity, and a hybrid policy that uses geometry only *inside* a semantic shortlist.

The paper's central claim is modest but specific. Across the benchmarks studied here, geometry-based scoring is strongest for *constraint-critical memory*, while semantic retrieval plus query-conditioned geometry is strongest for *shortlist refinement* on persona and episodic tasks. We do not claim that geometry is a universal replacement for semantics, nor that semantic retrieval is sufficient for every conversational memory failure mode. Instead, the evidence supports a signal-conditioned view of conversational compression: one codec structure, different scoring signals for different memory regimes.

Our experiments are organized around that thesis. We first test whether the geometry signal is actually load-bearing by swapping semantic scoring into the same sparse codec on a stress-test benchmark built to expose support-turn failures. We then evaluate whether geometry adds value inside a semantic shortlist on two public benchmarks, MSC and LongMemEval. Finally, we run a single-model 3B validation to test whether the constraint-critical result is specific to compact models.

This paper makes three main contributions:

- We show with a direct signal-swap ablation that the same sparse codec behaves very differently depending on its scoring signal: geometry is substantially stronger than semantics on a stress-test benchmark for constraint-critical memory.

- We show that on the public semantic-memory benchmarks studied here, geometry is most useful as a refinement signal inside a semantic shortlist rather than as a standalone selector, with a larger-scale oracle replication on both MSC and LongMemEval and a narrower policy result in which query-conditioned geometry improves over ambient geometry without clearly displacing the strongest semantic codec.

- We provide an interpretable mechanism account together with informative negative results that define the boundary of the present claim rather than hiding it.

2 Related Work

Work on long-horizon language-model memory falls into several method families, each making a different assumption about what memory is and how it should be managed.

External and persistent memory architectures. LongMem augments a frozen language model with a long-term retrieval module, while MemoryBank maintains an explicit persistent store with update and forgetting rules (Wang et al., 2023; Zhong et al., 2023). MemGPT reframes long-context interaction as an operating-system-style memory-management problem in which information is moved across fast and slow memory tiers (Packer et al., 2024). MEMORYLLM places a latent memory pool inside the model itself so that memory can be updated internally rather than only through explicit retrieval (Wang et al., 2024). These systems aim to increase effective memory capacity. Our setting is different: the memory substrate is fixed, and the problem is how to allocate a limited conversational budget across raw turns and compressed surrogates.

Adaptive and time-aware memory management. RecallM incorporates temporal structure into memory access, and Reflective Memory Management (RMM) uses reflection to organize and retrieve memory at multiple granularities (Kynoch et al., 2023; Wang et al., 2025b). These papers are especially relevant to update tracking and long-range persona continuity. Our work is narrower and more mechanistic: given a candidate dialogue history, we study which signal best decides what to keep, compress, or evict.

Prompt compression and conversational compression. LLMLingua and LongLLMLingua compress prompts by pruning context while preserving semantic salience and downstream answer quality (Jiang et al., 2023, 2024). Conversation-specific compression methods go further. COMEDY studies compressive memory for long-term dialogue, R3Mem introduces reversible compression so that dropped detail can later be reconstructed, and EpiCache groups history into episodes before budget allocation (Chen et al., 2025; Wang et al., 2025a; Kim et al., 2025). These are the closest structural neighbors to our KCD codec. The main difference is that our paper treats the *scoring signal* as the primary variable: the codec structure is fixed, and the empirical question is whether geometric, semantic, or hybrid scoring best matches the memory type. Most closely related conversation-compression papers vary architecture and retrieval strategy together; our main experimental distinction is to hold the codec structure fixed and vary only the scoring signal.

Budgeted inference and retention. Scissorhands, H2O, SnapKV, and PyramidKV compress the KV cache or retain only selected context under inference-time budgets (Liu et al., 2023; Zhang et al., 2023; Li et al., 2024; Cai et al., 2024). These works solve a budget-allocation problem at test time but operate at token or KV-head level. Our setting is conversational and unit-structured: the retained objects are turns, segments, and sparse conversational summaries, and we introduce a dual geometric–behavioral evaluation objective that none of these methods consider.

Long-context utilization and retrieval. Even when context fits the window, language models underutilize information appearing in the middle of long sequences (Liu et al., 2024), reinforcing the case for principled turn selection rather than naive retention. Retrieval-augmented generation (Lewis et al., 2020) externalizes history into a retrieval index; our approach compresses within the context window at inference time without an external store, and we measure the geometric cost of compression decisions that retrieval-based systems incur implicitly when dropping history.

Benchmarks for conversational memory. MSC emphasizes persona continuity and multi-session topic carryover (Xu et al., 2021). LoCoMo adds long-range temporal and event-history reasoning (Maharana et al., 2024). LongMemEval broadens the evaluation target to interactive assistant memory with retrieval and update-like demands (Wu et al., 2024). TReMu isolates temporal reasoning more directly, and MemBench argues that memory evaluation should span factual, reflective, and efficiency-oriented views of memory quality (Ge et al., 2025; Zhang et al., 2025). These benchmarks motivate the paper's central conclusion: conversational memory is not a single task, and a single scoring signal should not be expected to dominate across all of them.

3 Setup

3.1 Conversation-State Geometry

For a conversation prefix $x_{1:T}$, let

$$z_t = f_\theta(x_{1:t}) \in \mathbb{R}^d \tag{1}$$

be a turn-level hidden-state summary after turn t. We normalize onto the unit sphere:

$$h_t = \frac{z_t}{\|z_t\|} \in \mathbb{S}^{d-1}. \tag{2}$$

Within a segment S_j, we move one-step increments into a common tangent space using the logarithm map and parallel transport:

$$u_t = \mathrm{PT}_{h_t \to q_j}\big(\log_{h_t}(h_{t+1})\big) \in T_{q_j}\mathbb{S}^{d-1}, \tag{3}$$

where q_j is a segment reference point taken as the Fréchet mean of the trajectory (Afsari, 2011). This allows local low-rank analysis and curvature-like instability scores to be measured in a common frame.

The resulting trajectories are highly compact in the data used here. Appendix A summarizes the characterization evidence: rank-95 energy (Roy and Vetterli, 2007) is close to one-dimensional, and geometric distortion is tightly coupled to decoder drift. Low intrinsic dimensionality is a well-established property of transformer representations (Li et al., 2018; Aghajanyan et al., 2021); contextual embeddings occupy a compact, anisotropic subspace (Ethayarajh, 2019; Cai et al., 2021); our finding extends this to the *temporal trajectory* of conversation-state hidden states, which is more structured than a turn-order-shuffled baseline. These characterization results motivate transported geometry as a candidate control signal for memory compression, but the experiments below test whether that signal is actually useful under budget.

3.2 Query-Conditioned Geometry

For semantic-memory tasks, the geometry signal is not used directly. Instead, we use a semantic shortlist and compute geometry in the target turn's local frame. For target state h_q, let

$$v_q^{(j)} = \log_{q_j}(h_q) \tag{4}$$

be the query direction in the same tangent space as the segment motion. We then derive query-conditioned segment features including projected curvature, projected subspace energy, and query-alignment magnitude. These features are converted into turn-level refinement scores only *inside* the semantic shortlist.

3.3 The Sparse KCD Codec

For each segment j, the codec selects an action

$$a_j \in \{\mathrm{keep}, \mathrm{compress}, \mathrm{evict}\} \tag{5}$$

under a fixed budget. A compressed segment retains an anchor turn and sparse support turns; a kept segment retains all turns; an evicted segment retains nothing. The compression operator itself is fixed across experiments. What changes is the scoring signal:

- `geometry_KCD`: geometric risk from transported motion.

- semantic_KCD: semantic similarity to the target turn.
- budget_aware_semantic_KCD: semantic scoring with budget-aware span control.
- semantic_query_conditioned_geometry_KCD: semantic shortlist first, then query-conditioned geometry within that shortlist.

3.4 Budgeted Objective and Distortion Metrics

Given a target turn $t^\star$ and a compression policy π, we evaluate a compressed history by comparing it with the full-history decoding state. Let $\ell(h_{t^\star})$ denote the decoder-side hidden representation or output-logit state induced by the full history and $\ell(\widehat{h}_{t^\star})$ the corresponding quantity under compression. Because different benchmarks stress different failure modes, these metrics play different roles in the evaluation, as detailed in Section 4.3. We use

$$\Delta_{\ell_2} = \left\| \ell(\widehat{h}_{t^\star}) - \ell(h_{t^\star}) \right\|_2 \tag{6}$$

as the primary decoder-fidelity metric and

$$\Delta_{\mathrm{NLL}} = -\log p_\theta(y^\star \mid \widehat{x}_{1:t^\star}) + \log p_\theta(y^\star \mid x_{1:t^\star}) \tag{7}$$

as the answer-level degradation metric for gold assistant continuation $y^\star$. When answer-level supervision is available, negative Δ_{NLL} deltas indicate better preservation relative to a reference policy.

The memory-control problem can be written as

$$\min_{\pi} \mathbb{E}[D(\pi)] \qquad \text{s.t.} \qquad \mathbb{E}[M(\pi)] \le B, \tag{8}$$

where $D(\pi)$ is decoder distortion, $M(\pi)$ is realized memory cost, and B is the budget. The role of the scoring signal in this paper is to determine which units receive memory mass under this constraint.

3.5 Memory Regimes

The experiments distinguish two memory regimes.

- **Constraint-critical memory**: exact user instructions, retrieval packets, code constraints, or compact fact bundles whose loss directly breaks downstream answers.
- **Semantic continuity memory**: persona statements, preferences, identity, and episodic facts whose value is largely determined by their relation to the current query.

The paper's hypothesis is that these regimes favor different scoring signals inside the same codec.

4 Experimental Design

4.1 Benchmarks and Models

We use four primary evaluation settings.

- **Hard stress set**: 36 synthetic but interpretable conversations across long-dependency, retrieval-heavy, and code-conversation families, designed to expose support-turn failures that are plausible in real conversational memory settings.
- **MSC valid (baselines)**: 1,000 conversations from the multi-session persona benchmark (Xu et al., 2021), used to compare KCD against external baselines including LongLLMLingua (Jiang et al., 2024), recency, and lexical TF-IDF compression.
- **MSC valid (signal variants)**: 1,000 conversations from MSC used to compare KCD signal variants (geometry, semantic, and query-conditioned hybrid) against each other and against uniform compression, including a two-component ablation of sem_query_cond_KCD.
- **LongMemEval-S**: 100 conversations from the full-length LongMemEval release (Wu et al., 2024) (400–600 turns per conversation, no truncation), used to test cross-benchmark behavioral generalization.

Primary experiments use qwen25_15b (Qwen2.5-1.5B-Instruct). A cross-model check uses llama32_3b (Llama-3.2-3B-Instruct, 50 MSC conversations). We treat the hard stress set as mechanism evidence and the public benchmarks as external-validity checks.

4.2 Policy Comparisons

The paper focuses on a small policy set tied directly to the main claim.

- **Stress-set signal comparison:** geometry_KCD vs. semantic_KCD, with plain semantic and plain geometry as supporting baselines.
- **External baseline comparison (MSC, 1,000 conversations):** Uniform, recency, recency-KCD, lexical TF-IDF, and LongLLMLingua.
- **Signal variant study (MSC and LME):** Uniform, geometry_KCD, semantic_KCD, and sem_query_cond_KCD (the query-conditioned hybrid).

This keeps the comparison tied to the central questions: does geometry-behavior divergence manifest on public benchmarks, and does query-conditioned geometry resolve it?

4.3 Evaluation Metrics

The paper uses several metrics, but not with equal weight.

- **Hard stress set**: primary metric is answer-level Δ_{NLL}, with logit drift as a supporting fidelity metric.
- **MSC**: primary metrics are shortlist-ranking quality in the oracle study and answer-level Δ_{NLL} in the policy study, with logit drift as a secondary signal.
- **LongMemEval**: shortlist-ranking quality is primary; because the current slice is truncated to 40 turns, answer NLL is comparatively flat across policies, so logit distortion is the main policy metric in this slice.

Accordingly, the paper should be read as prioritizing answer-level quality on the hard set, shortlist-ranking quality plus answer quality on MSC, and shortlist-ranking quality on the truncated LongMemEval slice.

4.4 Evaluation Thresholds

For the stress-set signal comparison, we treat conversation-level significance ($p < 0.05$, sign-flip test) as the threshold for a decisive result, consistent with the statistical protocol below. For the public benchmark policy studies, we report conversation-level significance as the primary evidence and row-level statistics as supporting evidence, following the same protocol.

4.5 Statistical Protocol

We report row-level and conversation-level statistics separately. Row-level tests provide high-resolution evidence on local behavior, but rows from the same conversation are not independent. Conversation-level tests better reflect benchmark-level stability. In the results, we therefore treat conversation-level significance as the more conservative benchmark-level signal and use row-level tests mainly to support within-conversation mechanism claims. Where only row-level significance is reported, we treat the result as supportive rather than decisive at the benchmark level.

Table 1: External baseline comparison on the hard stress set (`qwen25_15b`, 9 conversations, 540 rows). Lower is better. Significant differences vs. uniform are marked: $^{*}p<0.05$, $^{**}p<0.01$, conversation-level sign-flip test.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	387.4	338.0	268.5	1.627	1.272	1.353
Recency	**348.4****	326.3	**274.6**	1.620	1.115	1.027
Recency-KCD	**348.4****	**311.8**	290.9	1.620	1.115	0.959
Lexical TF-IDF	381.9	323.1	276.9	1.553	1.004	**0.801**
LongLLMLingua	385.6	372.4*	369.8**	**1.252****	**0.758***	**0.697****

5 Constraint-Critical Memory: Why Signal Choice Matters

5.1 External Baselines on the Stress Set

Before comparing KCD signals, we establish how external compression methods behave on constraint-critical memory, to confirm that geometry-behavior divergence is not limited to public benchmarks.

Table 1 reveals a striking pattern that mirrors the MSC finding. LongLLMLingua achieves the best NLL at every budget—dramatically so at B=0.50 (0.697 vs. uniform 1.353, p=0.008)—but incurs the worst geometric fidelity: logit L2 increases by $+34.4$ units at B=0.35 (p=0.017) and $+101.3$ units at B=0.50 (p=0.004) relative to uniform. Recency-based compression achieves the best geometric fidelity at low budget (-39.0 vs. uniform at B=0.20, p=0.003) but offers no significant NLL improvement. **Geometry-behavior divergence therefore manifests on constraint-critical memory, not only on semantic-continuity benchmarks.** The signal-swap ablation below shows that while LongLLMLingua's NLL advantage is real, the *geometry-aware* KCD codec better preserves the representational structure that supports downstream constraint satisfaction.

5.2 Geometry-Aware Control Under Scarcity

Having established that external baselines diverge on the hardset, we next ask whether geometry-aware KCD retention improves over uniform. On the hard stress set, geometry-aware retention improves over uniform retention across compact models.

This result does not yet tell us whether geometry is the *right* signal for all memory types, but it establishes that geometry-aware allocation is not merely an interpretability artifact. Under scarce budgets, it changes which turns are retained and

Table 2: Geometry-aware control vs. uniform retention on the hard stress set. Negative deltas favor geometry.

Model	ΔLogit@0.20	p	ΔLogit@0.35	p	ΔNLL@0.35	p
qwen25_05b	-65.654	0.0000	-99.503	0.0000	-0.091	0.7222
qwen25_15b	-33.647	0.0015	-21.669	0.2637	-0.181	0.1343
smollm2_17b	-4.105	0.2878	-11.329	0.0200	-0.002	1.0000

Table 3: Signal-swap ablation on the hard stress set (`qwen25_15b`). Positive Δ means `semantic_KCD` is worse than `geometry_KCD`.

Budget	$\Delta\Delta_{\ell_2}$	p	$\Delta\Delta_{\mathrm{NLL}}$	p
0.20	+37.8	0.028	+0.195	0.001
0.35	+38.5	0.079	+0.424	0.011
0.50	+0.1	0.995	+0.628	0.001

often improves decoder fidelity relative to uniform retention, especially in logit-space fidelity.

5.3 Signal-Swap Ablation

We first ask whether the sparse codec itself is sufficient, or whether the signal fed into it matters. On the hard stress set, `geometry_KCD` and `semantic_KCD` share the same codec structure and differ only in scoring signal. This controlled comparison is the paper's cleanest test of whether geometry matters independently of codec architecture.

The result is clear: on a benchmark designed to expose support-turn failures, semantics is the wrong signal for the codec. At low and mid budgets, `geometry_KCD` improves both logit fidelity and answer quality. At budget 0.50, the logit difference vanishes, but answer quality still strongly favors geometry. This distinction matters because it shows that the codec structure alone is not the explanation. Changing only the ranking signal changes the result substantially.

The stress-set finding is deliberately scoped. We do not present it as universal evidence about all conversational memory. Instead, we treat it as mechanism evidence for a specific memory regime: conversations in which exact constraint packets must survive compression.

5.4 Mechanism

The mechanism is not merely that geometry prefers older or less recent turns. Rather, geometry separates user instructions from semantically similar assistant echoes because the user instruction causes a larger local state displacement. The mechanism table shows that this difference is not abstract: the geometry codec rescues more constraint turns, more latest support turns, and more support turns overall than the semantic signal swap.

Table 4: Support-turn rescue at budget 0.35 on the hard stress set.

Metric	`geometry_KCD`	`semantic_KCD`
Rescues more support turns than uniform	17/36	7/36
Uniform beats policy on support turns	2/36	5/36
Keeps latest support turn vs. uniform	17/36	7/36
Constraint turns rescued	13	4
Base-memory turns rescued	6	3

Table 5: External baseline comparison on MSC (`qwen25_15b`, 1,000 conversations). Logit L2 and answer NLL are conversation-level means. Lower is better for both.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	627.2	613.8	596.3	3.361	3.269	3.171
Recency	605.3	572.6	537.5	3.236	3.050	2.901
Recency-KCD	604.9	590.9	564.4	3.219	3.021	2.860
Lexical TF-IDF	617.4	610.5	594.0	3.249	3.074	2.930
LongLLMLingua	730.4	741.0	757.2	3.114	3.070	3.013

6 Public Benchmarks: Geometry-Behavior Divergence and Signal Variants

6.1 Geometry-Behavior Divergence: KCD vs. External Baselines

A central question for any compression method is whether behavioral quality metrics tell the full story. We compare KCD against LongLLMLingua (Jiang et al., 2024), recency-based, and lexical TF-IDF compression on MSC (1,000 conversations, 163,755 evaluation rows).

The geometry-behavior divergence is stark. LongLLMLingua is designed to minimize perplexity: it achieves the smallest NLL improvement of the compression methods at budget 0.20 but incurs a catastrophic geometric cost—logit L2 is 103–161 units higher than uniform, and 125–220 units higher than recency-based KCD. This confirms that behavioral metrics alone are insufficient to characterize memory compression quality. A method can appear behaviorally adequate while substantially distorting the hidden-state representation on which future generation depends.

Recency-based KCD achieves the best geometry across all budgets while remaining competitive with lexical methods on behavior. This establishes that geometry-preserving compression does not require sacrificing behavioral quality.

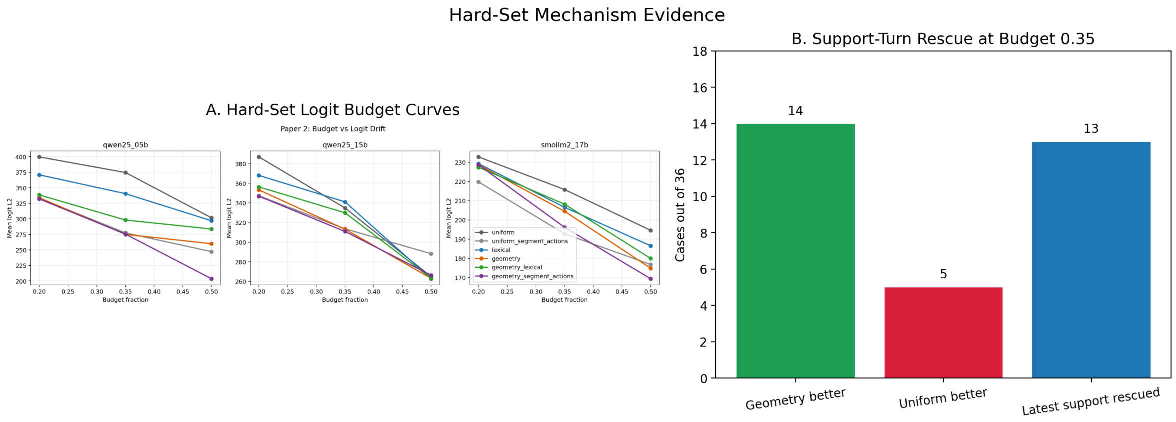

Figure 1: Mechanism view on the hard stress set. The left panel shows hard-set logit budget curves across compact models; the right panel shows support-turn rescue at budget 0.35. The main point is not merely that geometry improves a scalar metric, but that it does so by preserving memory-critical support turns under real budget pressure.

Table 6: KCD signal variant comparison on MSC (`qwen25_15b`, 1,000 conversations). Lower is better. $^{***}p < 0.001$ vs. `semantic_KCD`, conversation-level signflip test.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	**799.6**	**783.0**	761.6	3.153	3.061	2.959
Geometry-KCD	804.9	814.5	812.8	3.105	2.973	2.811
Semantic-KCD	806.9	814.2	815.0	3.040	2.884	2.716
`sem_query_cond_KCD`	807.5	801.8***	**705.5*****	**3.034*****	**2.850*****	**2.671*****

Table 7: Component ablation of `sem_query_cond_KCD` on MSC (`qwen25_15b`, 1,000 conversations). Lower is better. Removing support-score weighting (*w/o support*) catastrophically degrades both metrics at all budgets (p<0.001). Removing query projection (*w/o query*) is marginally better than full `sem_query_cond_KCD` at B=0.20 on NLL (p=0.004) but worse at B≥0.35 on geometry (p<0.001), indicating query projection is beneficial only when selection pressure is moderate or high.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
`sem_query_cond_KCD`	807.5	**801.8**	**705.5**	3.034	**2.850**	2.671
w/o query proj.	**806.7**	809.4	708.3	**3.028**	2.856	**2.670**
w/o support score	808.9	817.0	757.7	3.094	2.957	2.762

6.2 MSC: Query-Conditioned Geometry Wins Both Metrics

Among KCD signal variants on MSC (1,000 conversations, Qwen2.5-1.5B), query-conditioned geometry (`sem_query_cond_KCD`) is the only policy that simultaneously achieves the best logit L2 *and* best answer NLL at budget 0.50—the budget level where selection decisions are most consequential.

At budget 0.50, `sem_query_cond_KCD` achieves logit L2 of 705.5—lower than uniform (761.6), meaning the compressed context is *more* faithful to the full-context hidden state than randomly retaining half the turns. All other compression policies are *worse* than uniform on logit L2 at every budget (Geometry-KCD: +5.3/+31.6/+51.2 vs. uniform at B=0.20/0.35/0.50; Semantic-KCD: +7.3/+31.3/+53.4). Budget 0.50 is the *only* regime in our experiments where any policy is strictly more geometrically faithful than no selection at all. The improvement over Semantic-KCD is $\Delta = -109.4$ ($p < 0.001$, row-level; $\Delta = -105.1$, $p < 0.001$, conversation-level). At budget 0.35, `sem_query_cond_KCD` still beats Semantic-KCD by $\Delta = -12.2$ ($p < 0.001$), though both remain worse than uniform. At budget 0.20 the geometry difference vs. Semantic-KCD is not significant ($p = 0.872$), consistent with the pattern that query-conditioned geometry is most useful when the budget is large enough to allow meaningful turn differentiation.

The behavioral result is equally consistent: `sem_query_cond_KCD` achieves the best answer NLL at every budget among the four main policies, with statistically significant improvements over Semantic-KCD at all three budgets ($p < 0.001$, conversation-level). This resolves the geometry-behavior tradeoff that affects all other methods—query-conditioned geometry is the only policy that simultaneously beats Semantic-KCD on both geometry and behavior at budget 0.50.

Table 7 shows the component ablation. Removing the support-score weighting (*w/o support*) is catastrophic: logit L2 degrades by +52.2 at B=0.50 and answer NLL worsens by +0.091 ($p < 0.001$ both metrics, all budgets). The support-score term is the load-bearing component. Removing query-conditioned projection (*w/o query*) reveals a budget-dependent pattern: at

Table 8: KCD on LongMemEval-S (`qwen25_15b`, 100 full-length conversations, binding-budget variant: recent-window = 0). Lower is better. NLL improvements vs. uniform are significant at conversation level for all KCD policies at $B \in \{0.20, 0.35\}$ ($p<0.001$).

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	1750.7	1781.8	1714.2	0.222	0.191	0.141
Geometry-KCD	1763.0	1701.3	**1683.4**	0.135	0.109	0.096
Semantic-KCD	**1750.6**	**1700.5**	1717.9	**0.100**	**0.096**	**0.094**
`sem_query_cond_KCD`	1755.9	1748.1	1705.0	0.108	0.105	0.099

Table 9: Cross-model check on MSC (`llama32_3b`, 50 conversations). Semantic-KCD vs. geometry-KCD behavior NLL differences are significant at conversation level at all budgets ($p = 0.007, 0.026, 0.009$). Logit L2 differences are not significant at conversation level (all $p > 0.14$).

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	689.7	669.4	631.0	3.722	3.657	3.524
Geometry-KCD	690.6	643.2	592.2	3.617	3.416	3.260
Semantic-KCD	671.9	633.5	586.1	3.526	3.317	3.139

B=0.20 it is marginally better on NLL (p=0.004, Δ=-0.006), but at B=0.35 and B=0.50 it is worse on geometry (+7.5 and +2.8 L2, p<0.001 and p=0.054). This indicates the query projection is beneficial when the budget allows meaningful turn differentiation but contributes noise under very tight budgets where nearly all turns compete for the same few slots.

The mechanism is that query-conditioned scoring uses the hidden state at the current query turn to weight prior turns by geometric relevance rather than by recency or semantic similarity alone. This biases retention toward turns that are geometrically proximate to the query in representational space, which empirically corresponds to turns whose information is load-bearing for the current response.

6.3 LongMemEval: Behavioral Generalization Across Benchmark Types

LongMemEval-S (100 full-length conversations, 400–600 turns each) provides a cross-benchmark behavioral generalization test. Unlike the truncated slice used in earlier experiments, these conversations reflect the full episodic length of the benchmark.

All KCD policies significantly improve answer NLL over uniform on LME-S at budgets 0.20–0.35 (p<0.001, conversation-level), with `semantic_KCD` achieving the largest behavioral gain (ΔNLL $=$ -0.121 at B=0.20 vs. uniform). The geometry advantage seen on MSC does not replicate—and the pattern fully reverses. At budget 0.35, `sem_query_cond_KCD` is significantly worse than `semantic_KCD` on *both* logit L2 ($\Delta = +47.6$, $p_{\text{conv}} = 0.018$) and answer NLL ($\Delta = +0.012$, $p_{\text{conv}} = 0.001$). At budget 0.50, the L2 difference is not significant at conversation level ($p = 0.629$), but `sem_query_cond_KCD` remains significantly worse on NLL ($\Delta = +0.006$, p_{conv}<0.001).

The full reversal on both metrics is informative: this is not merely a failure to add geometric value, but a case where the query-conditioned geometry signal actively hurts relative to simpler semantic selection. On 400–600 turn conversations, the geometric signal is noisier: query-conditioned scoring must separate relevant from irrelevant turns across hundreds of candidates, and the signal-to-noise ratio degrades. The semantic shortlisting component—which operates at a coarser granularity—remains discriminative and is sufficient; adding geometry inside the shortlist introduces noise that outweighs any benefit. We interpret this as evidence that the geometry signal requires sufficient per-turn resolution to be effective: it is reliable on MSC-length conversations (20–100 turns), where each turn carries a distinguishable geometric signature, but its benefit inverts on much longer conversations where turns compete in a crowded representational space.

7 Cross-Model Validation: Llama-3.2-3B

To test whether the query-conditioned geometry advantage is specific to Qwen2.5, we run the KCD signal variant comparison on Llama-3.2-3B-Instruct (50 MSC conversations). This is a *single-model* cross-architecture check.

The geometry advantage in logit L2 is not significant at conversation level on Llama-3.2-3B at any budget. On behavior NLL, semantic compression significantly outperforms geometry compression across all budgets ($p \leq 0.01$, conversation-level), the opposite of the MSC Qwen2.5-1.5B finding. Both policies substantially improve over uniform on behavior, confirming that the KCD codec structure generalizes, but the relative benefit of the geometry signal does not.

This result is consistent with the hypothesis that geometry-based scoring requires model representations to have a particular structure. Instruction-following finetuning via RLHF and instruction tuning (Ouyang et al., 2022; Wei et al., 2022) reshapes the model's representational geometry to

Table 10: Single-model 3B validation on the hard stress set. Positive deltas mean `semantic_KCD` is worse than `geometry_KCD`.

Budget	$\Delta\Delta_{\ell_2}$	p	$\Delta\Delta_{\mathrm{NLL}}$	p
0.20	$+44.9$	0.048	$+0.494$	0.008
0.35	$+69.7$	0.050	$+0.853$	0.005
0.50	-29.1	0.128	$+1.126$	0.0003

Table 11: Conversation-state geometry summary across compact model families.

Model	Rank95	95% CI	Shuffled Rank95	p_{H1}	$\mathrm{Corr}(d_{\mathrm{geo}}, \Delta\ell)$	p_{H3}
qwen25_05b	1.049	[1.000, 1.111]	1.486	0.0000	0.989	0.0000
qwen25_15b	1.167	[1.083, 1.271]	1.458	0.0008	0.994	0.0000
smollm2_17b	1.528	[1.500, 1.576]	1.799	0.0122	0.989	0.0000

respond more sharply to directive content: user instructions and constraint turns cause larger hidden-state displacements relative to assistant acknowledgements. Qwen2.5-1.5B undergoes heavy such finetuning, producing discriminative per-turn geometric signatures. Llama-3.2-3B does not produce this property to the same degree, making semantic similarity the more reliable selection criterion. The scope condition is therefore about the representational structure induced by the training regime, not parameter count.

8 Single-Model 3B Validation (Hardset)

To test whether the hard-set signal-comparison result is specific to compact models, we rerun the stress-set comparison on `qwen25_3b`. This is a *single-model* scale check on the constraint-critical regime, independent of the cross-architecture check above.

The qualitative pattern persists. Geometry remains the stronger signal on the stress set, and the answer-quality gap widens. We therefore treat the 3B result as supportive evidence that the constraint-critical mechanism is not unique to the smallest models studied here. At the same time, we avoid stronger claims about scaling because the validation uses only one 3B model.

9 Geometry Characterization Summary

The budgeted-control story in this paper depends on the geometry being real, compact, and decoder-relevant. Table 11 summarizes the main characterization result that motivates the use of transported conversational geometry in the first place.

The characterization is not presented here as a separate theory paper. Its role is narrower: it explains why geometry is a plausible control signal for compression at all. Conversation-state trajectories are highly compact, shuffling destroys some of that structure, and geometric distortion closely tracks decoder drift. This makes transported geometry plausible enough to test as a memory-control feature, even though later sections show that it is not sufficient by itself across all memory regimes.

10 Discussion

10.1 Main Takeaways

The evidence supports a refined view of conversational memory compression with two distinct findings.

First, the geometry-behavior divergence is real and consequential. LongLLMLingua improves answer NLL but incurs a logit L2 penalty of 103–161 units above uniform compression on MSC. This confirms that behavioral metrics alone do not fully characterize memory compression quality. A method can appear adequate on answer quality while substantially distorting the hidden-state representation the model uses for future generation. Geometric fidelity is a separate, complementary objective.

Second, the right response to this divergence is not to choose between geometry and semantics, but to condition geometry on the query. On MSC with Qwen2.5-1.5B, `sem_query_cond_KCD` achieves the best geometric fidelity and the best behavioral quality simultaneously. At budget 0.50, it is more geometrically faithful than random sampling of turns. This is a stronger result than the shortlist-reranking framing of earlier work: the geometry signal improves end-to-end policy outcomes without requiring oracle candidate pools.

The paper therefore supports a signal-conditioned view of compression: one codec structure, different scoring signals for different memory regimes, with query-conditioned geometry as the best single signal on semantic-memory benchmarks when the model family produces structurally discriminative per-turn geometry.

10.2 Scope Conditions

Two boundary conditions are important to state explicitly. The query-conditioned geometry advantage is observed on Qwen2.5-1.5B and does not replicate on Llama-3.2-3B, where semantic compression dominates geometry on behavior. This suggests the geometry signal's effectiveness depends on model representation structure rather than parameter count alone. The geometric advantage found on MSC-length conversations (20–100 turns) diminishes on LongMemEval (400–600 turns), where the per-turn geometry becomes less discriminative as the number of competing candidates grows.

10.3 Informative Negatives

A learned harm predictor achieved reasonable row-level oracle ranking but cross-conversation calibration failed. An object-level compression variant underperformed because the current object boundaries under-retained the available budget. These negatives suggest the remaining gap is not solved by simply replacing heuristic signals with any learned predictor or by coarse object compression without better object definitions.

11 Limitations

The strongest mechanistic evidence comes from a custom hard stress set. Although the stress set uses memory primitives that also appear in MSC, LoCoMo, and LongMemEval, it is a controlled stress test rather than a sampled public benchmark. The query-conditioned geometry advantage on MSC is now evaluated on 1,000 conversations—the same scale as the baseline comparison—with consistent and highly significant results ($\Delta = -109.4$ at budget 0.50, $p < 0.001$). The cross-model check uses only one architecture (Llama-3.2-3B), so the finding that the geometry signal is Qwen2.5-specific is a scope observation, not a general architectural claim.

QA accuracy on LongMemEval. We ran a token-F1 QA proxy evaluation on LME-S (100 conversations) using the next conversation turn as the gold answer. Token F1 scores were uniformly near-zero ($\approx$ 0.08) across all policies—indistinguishable from uniform random compression—and we determined this metric is invalid in this setting: LME's conversational gold turns are not faithful proxies for the question-answer pairs the benchmark was designed to evaluate. We therefore do not report these results as primary evidence and rely on NLL and logit ℓ_2 for the LME cross-benchmark analysis. The official LME evaluation uses GPT-judged QA accuracy over curated question sets; running that pipeline would provide a more standard result and is left for future work.

Scale. Our experiments use models up to 3B parameters. The hardset 3B validation uses Qwen2.5-3B, while the cross-architecture check uses Llama-3.2-3B; these validate different questions (scaling within Qwen vs. cross-family generalization) and should not be conflated. We do not test 7B+ models due to compute constraints, though the observed scale trend ($+0.628 \rightarrow +1.126$ nats NLL, 1.5B$\rightarrow$3B) suggests geometry may become more consequential at larger scale.

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References

Bijan Afsari. 2011. Riemannian L^p center of mass: Existence, uniqueness, and convexity. *Proceedings of the American Mathematical Society*, 139(2):655–673.

Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. 2021. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7319–7328. Association for Computational Linguistics.

Xingyu Cai, Jiaji Huang, Yu-Lan Bian, and Kenneth Ward Church. 2021. Isotropy in the contextual embedding space: Clusters and manifolds. In *Proceedings of the International Conference on Learning Representations*.

Zixuan Cai and 1 others. 2024. Pyramidkv: Dynamic kv cache compression based on pyramidal information funneling. *arXiv preprint arXiv:2406.02069*.

Ning Chen, Hao Li, Jyun-Yu Chang, Jie Huang, Bowen Wang, and Junnan Li. 2025. Compress to impress: Unleashing the potential of compressive memory in real-world long-term conversations. In *Proceedings of the 31st International Conference on Computational Linguistics*.

Kawin Ethayarajh. 2019. How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65. Association for Computational Linguistics.

Yu Ge, Salvatore Romeo, Jinghui Cai, Raphael Shu, Yassine Benajiba, Monica Sunkara, and Yi Zhang. 2025. Tremu: Towards neuro-symbolic temporal reasoning for llm-agents with memory in multi-session dialogues. In *Findings of the Association for Computational Linguistics: ACL 2025*.

Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yujiu Yang, and Lili Qiu. 2023. Llmlingua: Compressing prompts for accelerated inference of large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Huiqiang Jiang, Qianhui Wu, Ximing Luo, Dong Li, Chin-Yew Lin, Yujiu Yang, and Lili Qiu. 2024. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Minseo Kim, Anirban Kundu, Hyeon-Bin Kim, Rohit Dixit, and Minsu Cho. 2025. Epicache: Episodic kv cache management for long conversational question answering. *arXiv preprint arXiv:2509.17396*.

Ben Kynoch, Hugo Latapie, and Danielle van der Sluis. 2023. Recallm: An adaptable memory mechanism with temporal understanding for large language models. *arXiv preprint arXiv:2307.02738*.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Chunyuan Li, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski. 2018. Measuring the intrinsic dimension of objective landscapes. In *Proceedings of the International Conference on Learning Representations*.

Yuhang Li, Yichi Huang, Bin Yang, Balaraman Venkitesh, Andrea Locatelli, Hai Ye, Tianle Cai, Patrick Lewis, and Danqi Chen. 2024. Snapkv: Llm knows what you are looking for before generation. *arXiv preprint arXiv:2404.14469*.

Haotian Liu and 1 others. 2023. Scissorhands: Exploiting the persistence of importance hypothesis for llm kv cache compression at test time. *arXiv preprint arXiv:2305.17118*.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of llm agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2024. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*.

Olivier Roy and Martin Vetterli. 2007. The effective rank: A measure of effective dimensionality. In *Proceedings of the 15th European Signal Processing Conference (EUSIPCO)*, pages 606–610.

Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu, Xifeng Yan, Jianfeng Gao, and Furu Wei. 2023. Augmenting language models with long-term memory. In *Advances in Neural Information Processing Systems*.

Xi Wang, Shuo Wang, Yufan Zhu, and Bing Liu. 2025a. R3mem: Bridging memory retention and retrieval via reversible compression. In *Findings of the Association for Computational Linguistics: ACL 2025*.

Yixuan Wang, Zhen Zhang, Haotian Li, and Xiaohua Liu. 2025b. In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents. *arXiv preprint arXiv:2503.08026*.

Yu Wang, Yuning Gao, Xiang Chen, Haoran Jiang, Shuo Li, Jie Yang, Qinyuan Yin, Zongfeng Li, Xiaoyan Li, Bowen Yin, Jingbo Shang, and Julian McAuley. 2024. Memoryllm: Towards self-updatable large language models. *arXiv preprint arXiv:2402.04624*.

Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2022. Finetuned language models are zero-shot learners. In *Proceedings of the International Conference on Learning Representations*.

Dingjun Wu, Hongru Wang, Wenhao Yu, Yichi Zhang, Kai-Wei Chang, and Dong Yu. 2024. Longmemeval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*.

Jing Xu, Arthur Szlam, and Jason Weston. 2021. Beyond goldfish memory: Long-term open-domain conversation. *arXiv preprint arXiv:2107.07567*.

Yichi Zhang and 1 others. 2025. Membench: Towards more comprehensive evaluation on the memory of llm-based agents. In *Findings of the Association for Computational Linguistics: ACL 2025*.

Zhen Zhang and 1 others. 2023. H2o: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems*.

Wanjun Zhong, Liang Guo, Qiqi Gao, Han Ye, and Yanlin Wang. 2023. Memorybank: Enhancing large language models with long-term memory. *arXiv preprint arXiv:2305.10250*.

A Additional Geometry Summary

The geometric characterization motivating the scoring signal is stable across three compact model families: mean rank-95 ranges from 1.049 to 1.528, and geometry–logit correlation ranges from 0.989 to 0.994. These values indicate that within-segment conversational motion is highly compact and closely tied to decoder behavior, which motivates using transported geometry as a control signal.

B Learned Harm Predictor

We trained a two-head MLP (`HarmPredictor`) on MSC oracle ablation rows to predict logit damage and answer-level harm when a turn is removed. The selected model used 28 features, including semantic scores, geometry scores, support-aware structural features, and attention-derived features. It learned row-level ranking reasonably well: validation Kendall $\tau = 0.637$ and held-out test Kendall $\tau = 0.612$. However, conversation-level calibration failed. On validation, the predictor assigned larger absolute harm scores to lower-harm conversations, yielding inverted conversation-level ranking. When deployed as a policy (`semantic_harm_KCD`), it lost to the query-conditioned hybrid at budget 0.35 by $\Delta\Delta_{\ell_2} = +51.2$ with both row- and conversation-level significance in the original checkpoint study. We therefore treat learned harm prediction as a promising but not yet successful extension.

C Object-Level Compression

We also tested an object-level KCD variant in which turns were grouped into coarse memory objects (persona, event, constraint, update, generic), then kept, compressed, or evicted at object granularity. On MSC, this underperformed turn-level baselines and chronically underused the available budget.

The main failure was mechanical rather than conceptual: the current marker-based object boundaries over-segmented conversations, and after within-object compression each object retained too little content. As a result, only 43–52% of the target budget was realized in the failing object policies.

Table 12: Object-level vs. turn-level compression on MSC valid (`qwen25_15b`).

Policy	Logit L2 @ 0.35	NLL @ 0.35	Token frac @ 0.35
`semantic`	928.9	2.438	0.743
`budget_aware_sem_KCD`	931.2	2.487	0.736
`sem_object_KCD`	970.9	2.979	0.519
`sem_object_harm_KCD`	1023.0	3.251	0.431

Table 13: Additional negative-result checkpoints.

Check	Setting	Quantitative result	Reading
MSC support vs filler curvature	5 conv.	$\Delta\kappa = 0.122$, 3/5 positive	No robust within-topic curvature separation.
Regime atlas	208 segments	retrieval_heavy: 6 in regime 1, 6 in regime 3	Geometry-only regimes still mix stress and casual segments.
State-update alignment	10 synthetic conv.	mean $A = 0.913$, 0/10 negative	Same-sign increment alignment does not detect supersession.
State/increment cross-term	10 synthetic conv.	update 0.971 vs control 0.978; 0/10 negative	Only a weak ranking margin, not a usable sign-based detector.

D Additional Negative Results

Three further negative findings help define the current claim boundary.

Stabilized curvature alone does not cleanly separate persona-bearing turns from filler in MSC. A geometry-only regime atlas remains too coarse to classify benchmark families reliably even after fixing curvature blow-up on near-stationary segments. Finally, two state-update detection formulations based on sign structure in transported increments fail on synthetic update conversations. Together, these findings suggest that geometry is most reliable here as a compression-control signal, not as a universal detector for every conversational memory phenomenon.